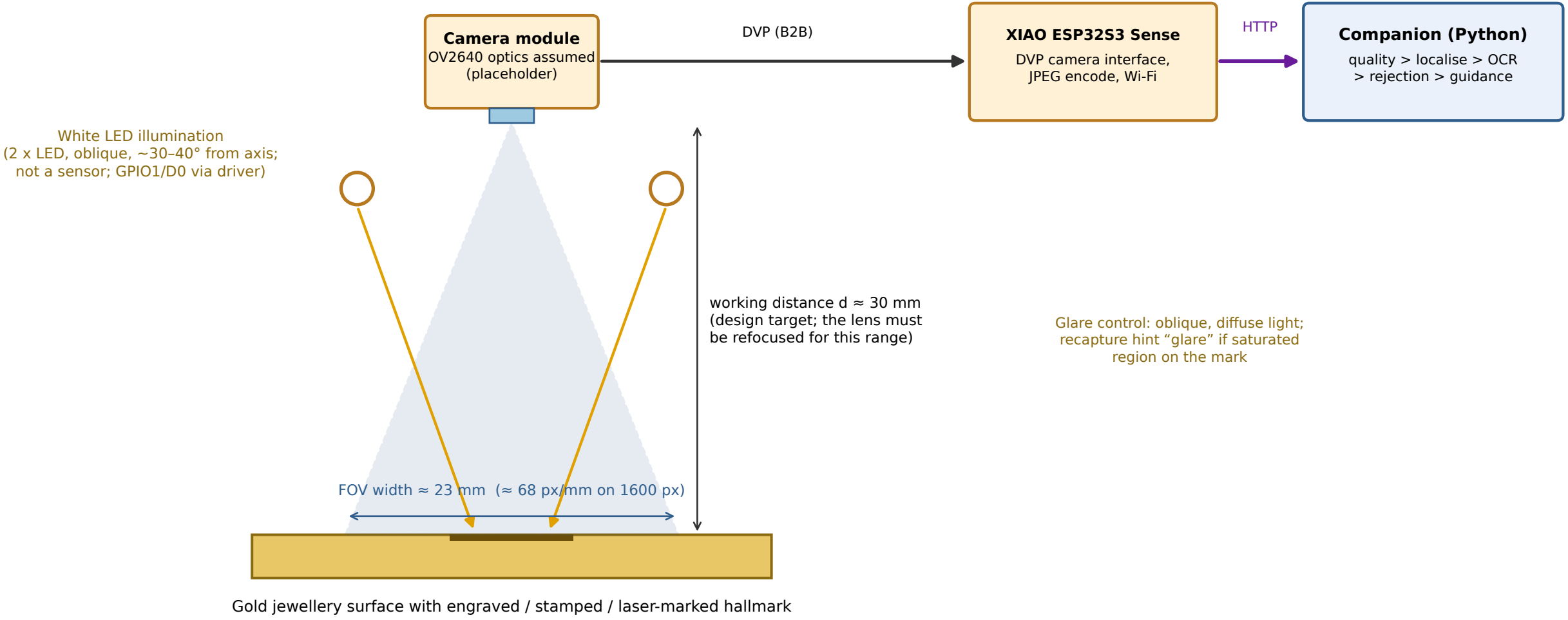
Camera and optical sensing schematic (PROPOSED design, not built, not simulated)

Status: design estimate only. Wokwi cannot simulate a camera, so experiments use synthetic images in place of this sensor.
Proposed module: OV5640 AF (the vendor says kits may ship with an OV5640 or OV3660, so the fitted module is unconfirmed); no field of view or minimum focus distance is listed, so they cannot be estimated here. The cone and the 23 mm figure below use OV2640 numbers (Arducam page: 1600x1200, fixed focus, EFL 4.22 mm, 52 deg diagonal FOV) as an illustrative PLACEHOLDER, not OV5640 values, and all optical values are estimates; macro focus must be measured on the delivered module (refocus, or add a close-up lens) before any build.
Fallback architecture: USB macro camera on the companion computer (no image transfer from the wrist unit).



Example: a 1.5 mm character is ≈ 103 px tall at d = 30 mm; the synthetic images used in the experiments are 640x400 px, i.e. a lower resolution than this design.